

PGR210 – NLP part

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Outline

- Practical applications
- NLP Pipeline with a simple chatbot example

2.1. Practical applications

- Natural language processing is everywhere.

Table 1.1 Categorized NLP applications

Search	Web	Documents	Autocomplete
Editing	Spelling	Grammar	Style
Dialog	Chatbot	Assistant	Scheduling
Writing	Index	Concordance	Table of contents
Email	Spam filter	Classification	Prioritization
Text mining	Summarization	Knowledge extraction	Medical diagnoses
Law	Legal inference	Precedent search	Subpoena classification
News	Event detection	Fact checking	Headline composition
Attribution	Plagiarism detection	Literary forensics	Style coaching
Sentiment analysis	Community morale monitoring	Product review triage	Customer care
Behavior prediction	Finance	Election forecasting	Marketing
Creative writing	Movie scripts	Poetry	Song lyrics

GROUP PRESENTATION:
Have you heard of all these applications?

Pick one application that sounds most unfamiliar to you, and seek more introduction information to prepare a group presentation.

2.1.1. Examples of tasks

Easy

- Spell checking
- Keyword search
- Finding synonyms

Medium

- Parsing information from websites, documents

Hard

- Machine translation
- Semantic analysis
- Question answering

2.1.2. Representative application on hard ones

- Machine translation

<http://demo.statmt.org/index.cgi>

2.1.3 Representative application on hard ones

- Semantic analysis

*Recent progress on Transforme



ChatGPT4 does not currently exist as it is a hypothetical model, but I can explain how the transformer and attention models have contributed to the development of the previous versions of GPT models.



The transformer model is a type of neural network architecture that was introduced in the paper "Attention is All You Need" by Vaswani et al. (2017). It has revolutionized the field of NLP by providing an alternative to the traditional sequence-to-sequence models, such as the LSTM and GRU models. The transformer model is based on a self-attention mechanism that allows the model to focus on the most relevant parts of the input sequence at each step of the computation. This makes it more efficient and effective at capturing long-range dependencies in the input sequence.

The transformer architecture has been used in several language models, including GPT-1, GPT-2, and GPT-3, which are large-scale pre-trained language models developed by OpenAI. These models have achieved state-of-the-art performance on several NLP benchmarks, such as language modeling, text generation, and question answering.

In particular, the attention mechanism has enabled these models to capture complex linguistic structures and relationships between words and sentences, which has improved their ability to generate coherent and contextually appropriate responses in conversational settings. This has made them particularly useful for developing conversational AI

2.1.3 Representative application on hard ones

- Question answering

https://colab.research.google.com/github/spark-ming/albert-qa-demo/blob/master/Question_Answering_with_ALBERT.ipynb#scrollTo=rkivu8FOqp_8

2.1.3 Representative application on hard ones

- Text to speech

https://cloud.google.com/text-to-speech/?utm_source=google&utm_medium=cpc&utm_campaign=emea-no-all-en-dr-bkws-all-all-trial-e-gcp-1010042&utm_content=text-ad-none-any-DEV_c-CRE_495029614056-ADGP_Hybrid%20%7C%20BKWS%20-%20EXA%20%7C%20Txt%20~%20AI%20%26%20ML%20~%20Text-to-Speech%23v4-KWID_43700060416646650-aud-606988878614%3Akwd-477477965678-userloc_1011021&utm_term=KW_text%20to%20speech%20google-NET_g-PLAC_&gclid=Cj0KCQjw0caCBhCIARIsAGAfzMzk25tynK3sSblfA1rrvsThxcKopx30fjChRQgjH9s54EnN6jll5vcaAoGVEALw_wcB&gclsrc=aw.ds

2.1.4. A representative application

- 2018, Google, Wavenet:

<https://www.youtube.com/watch?v=JjK8apEishQ&t=175s>

<https://deepmind.com/blog/article/wavenet-generative-model-raw-audio>

- Clone voices:

https://www.youtube.com/watch?v=-O_hYhToKoA

Steps:

git clone <https://github.com/CorentinJ/Real-Time-Voice-Cloning.git>

Demos

- Fake voice text to speech using deep learning:

<https://www.youtube.com/watch?v=6bFN2YkN6bo>

2.2. NLP Pipeline

Outline

- 1. A simple chatbot example
- 2. A chatbot natural language pipeline

Outline

- ***1. A simple chatbot example***
- 2. A chatbot natural language pipeline

1. A simple chatbot example

User	Hello	Hi	Good morning	Hi Rosa	Hihi
Chatbot	Hello	Hello	Good morning	Hello	

1. An simple chatbot example

- '|' means “OR”
- '*' means the preceding character can occur 0 or more times and still match.

```
r = "(hi|hello|hey) [ ]* ([a-z]*)"
```

specify a character class with square brackets

The star ('*') after a character class means that the regular expression will match any number of consecutive characters if they are all within that character class

the regular expression "[a-z]" will match any single lowercase letter, “a” through “z.”

match greetings that start with “hi” or “hello” or “hey” followed by any number of '<space>' characters and then any number of letters.

What is regular expressions?

- Regular expressions use a special kind (class) of formal language grammar called a regular grammar
- Regular grammars have predictable, provable behavior, and yet are flexible enough to power some of the most sophisticated dialog engines and chatbot on the market
- Deep, complex regular grammar rules can often be expressed in a single line of code called a regular expression
 - Amazon Echo, Google Home, and similarly complex and useful assistants use this kind of language to encode the logic for most of their user interaction

What is regular expressions?

- There are two “official” regular expression packages in Python
- We use the re package here just because it’s installed with all versions of Python
- The regex package comes with later versions of Python and is much more powerful”

A regular expression (or RE) specifies a set of strings that matches it; the functions in this module let you check if a particular string matches a given regular expression (or if a given regular expression matches a particular string, which comes down to the same thing).

<https://docs.python.org/3/library/re.html>

Outline

- 1. A simple chatbot example
- ***2. A chatbot natural language pipeline***

2. A chatbot natural language pipeline

- A chatbot requires four kinds of processing as well as a database to maintain a memory of past statements and responses.

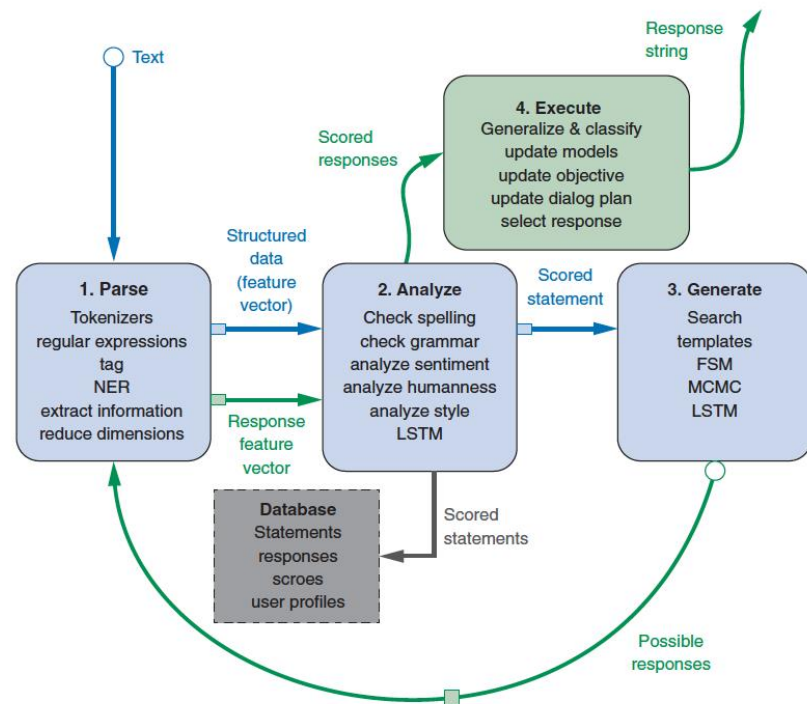


Figure 1.3 Chatbot recirculating (recurrent) pipeline

2. A chatbot natural language pipeline

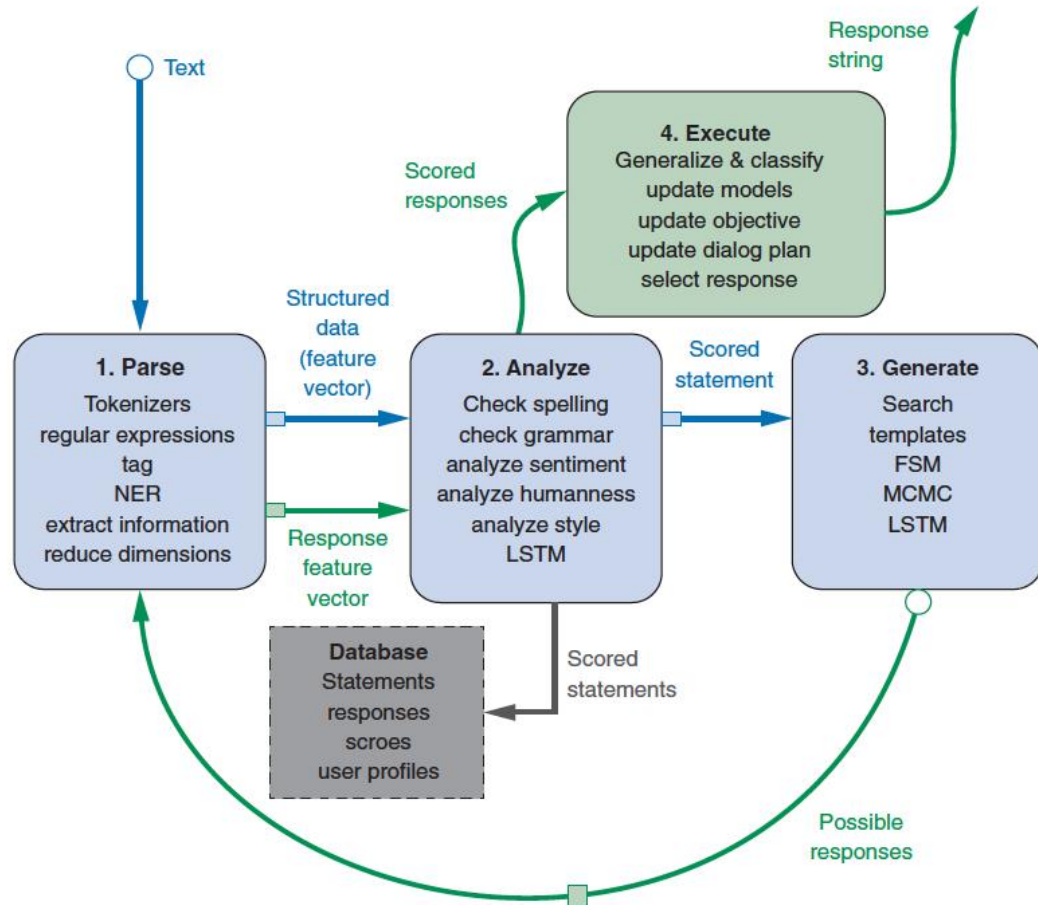


Figure 1.3 Chatbot recirculating (recurrent) pipeline

1. Parse—Extract features, structured numerical data, from natural language text.

2. Analyze—Generate and combine features by scoring text for sentiment, grammaticality, and semantics.

3. Generate—Compose possible responses using templates, search, or language models.

4. Execute—Plan statements based on conversation history and objectives, and select the next response.

- How to build a general NLP application?
 - Normally walk through the requirements
 - Break the problem down into several sub-problems
 - Try to develop a step-by-step procedure to solve them

This step-by-step processing of text is known as a *pipeline*

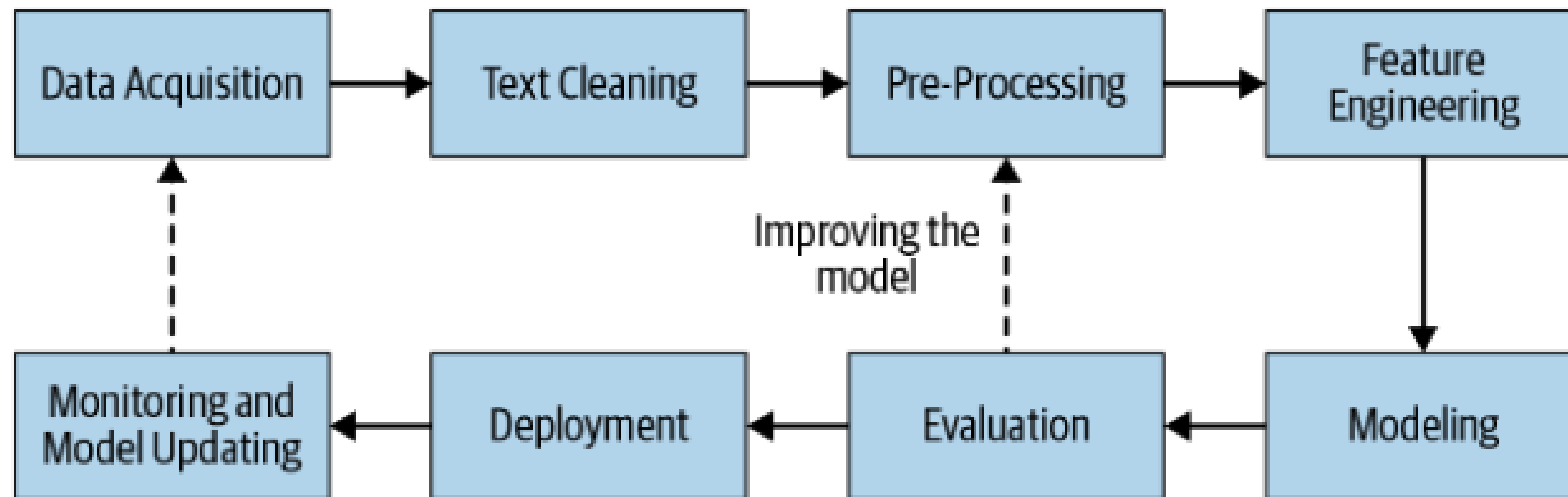


Figure 2-1. Generic NLP pipeline

- Note that:
 - exact step-by-step procedures may depend on the specific task at hand. For example, a text-classification system may require a *different* feature extraction step compared to a text-summarization system.

Discussion on the pipeline of an NLP application

- We need to develop an NLP system to identify whether an incoming customer query (for example, using a chat interface) is a sales inquiry or a customer care inquiry.
- How can one go about building such a system? Discuss on the factors that need to be considered and steps of developing such a system (pipeline)

Data Acquisition

- In an ideal setting, we'll have the required datasets with thousands—maybe even millions—of data points.
- For example, we have historic queries from previous years, which sales and support teams responded to. Further, the teams tagged these queries as sales, support, or other. So, not only do we have the data, but we also have the labels.

Text Extraction and Cleanup

- Text extraction and cleanup refers to the process of extracting raw text from the input data by removing all the other non-textual information, such as markup, metadata, etc., and converting the text to the required encoding format.